Prob - 1

07 July 2026 11:56 PM

3 Questions

Task 1: 4-Wheel Independent Steering Geometry

You are given a 4-wheel independent drive and steer rover. The robot has to rotate about a centre point O located somewhere in the robot's local plane.

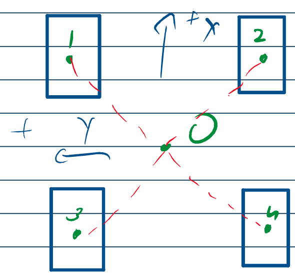
Given:

- **Wheel Positions:**
  - Front Left (FL): (0.2 m, 0.2 m)
  - Front Right (FR): (0.2 m, -0.2 m)
  - Rear Left (RL): (-0.2 m, 0.2 m)
  - Rear Right (RR): (-0.2 m, -0.2 m)
- **Max Steering Range:** $\pm 90°$
- **Wheel Radius:** 0.08 m

Problem Statement:
When the rover needs to rotate about a point O = (x_o, y_o), compute the steering angle θ_i for each wheel. **Solve for the following cases:**

Compute the steering angles for the following centres of rotation.

- **Case 1:** Rotate about **O = (0,0)** (on-spot rotation for the rover)



Point a = (x_a, y_a)

For rover to rotate about O, ICR must be O

∴ For ICR to be O, wheels should be geometrically $\perp$ to line joining a & O

$$\therefore \boxed{\theta_i = \pm 90 + \tan^{-1}(\text{slope of } a - O)}$$

For Case I :-

pt $(0.2, 0.2)$ & $(0,0)$ $\rightarrow$ slope $= 1$, angle $= 45^\circ$

$\therefore$ $\boxed{\text{wheel angle} = -90 + 45 = -45}$

pt $(-0.2, 0.2)$ & $(0,0)$ $\rightarrow$ slope $= -1$, angle $= 135^\circ$

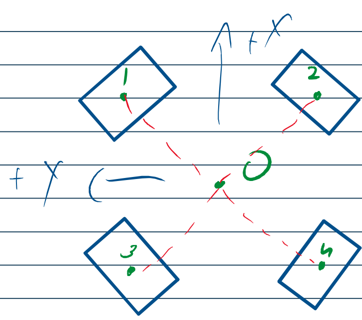
$\therefore$ $\boxed{\text{wheel angle} = -90 + 135 = 45^\circ}$

pt $(-0.2, -0.2)$ & $(0,0)$ $\rightarrow$ slope $= 1$, angle $= -135$

$\therefore$ $\boxed{\text{wheel angle} = +90 - 135 = -45^\circ}$

pt $(0.2, -0.2)$ & $(0,0)$ $\rightarrow$ slope $= -1$, angle $= -45$

$\therefore$ $\boxed{\text{wheel angle} = +90 - 45 = 45^\circ}$



- **Case 2:** Rotate about $\mathbf{O} = \mathbf{(1,0)}$ (1m to the right of the rover)

pt $(0.2, 0.2)$ & $(1,0)$, slope $= -0.25$, angle ≈ 165.96

pt $(0.2, 0.2)$ & $(1,0)$, slope $= -0.25$, angle ≈ 165.96

$$\boxed{\therefore \text{wheel angle} = -90 + 165.96 = 75.96}$$

pt $(0.2, -0.2)$ & $(1,0)$, slope $= 0.25$, angle ≈ -165.96

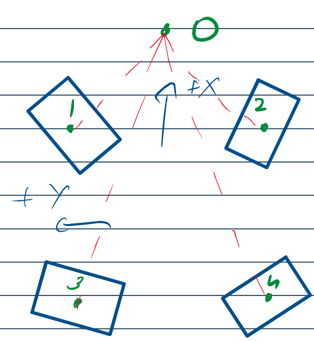
$$\boxed{\therefore \text{wheel angle} = +90 - 165.96 = -75.96}$$

pt $(-0.2, 0.2)$ & $(1,0)$, slope $= -0.166$, angle ≈ 170.54

$$\boxed{\therefore \text{wheel angle} = -90 + 170.54 = 80.54}$$

pt $(-0.2, -0.2)$ & $(1,0)$, slope $= 0.166$, angle ≈ -170.54

$$\boxed{\therefore \text{wheel angle} = -90 + 170.54 = -80.54}$$



- **Case 3:** Rotate about **O = (-2,0)** (2m to the left of the rover)

For pt $(0.2, 0.2)$ & O, slope $= 0.09$, angle $= 5.19°$

$$\boxed{\therefore \text{wheel angle} = -90 + 5.19 = -84.81°}$$

For pt $(0.2, -0.2)$ & O, slope $= -0.09$, angle $= -5.19°$

$$\boxed{\therefore \text{wheel angle} = +90 - 5.19 = 84.81°}$$

For pt $(-0.2, 0.2)$ & O, slope $= 0.11$, angle $= 6.34$

$$\boxed{\therefore \text{wheel angle} = -90 + 6.34 = -83.66°}$$

For pt $(-0.2, -0.2)$ & O, slope $= -0.11$, angle $= -6.34$

$$\boxed{\therefore \text{wheel angle} = +90 - 6.34 = 83.66°}$$